

Annotated State Traces and Routing Accuracy Table

Trace 1: Prediction

1. Query

Who will win Pies vs Cats?

2. Router Decision

- **Intent:** prediction
- **Route:** prediction

3. Tool Called: `match_winner_prediction_tool`

- **Input:**
 - `team_a` = Collingwood Magpies
 - `team_b` = Geelong Cats
- Match Date: 2025-05-03
- Predicted Winner: Collingwood Magpies
- Probability: 62.7%
- Away Win: 36.7%
- Draw: 0.6%
- Home Win: 62.7%

4. Validation

- **Status:** valid
- **Reason:** Tool result passed validation.

5. Final Response

Model prediction for Collingwood Magpies vs Geelong Cats on 2025-05-03:

Predicted winner: Collingwood Magpies

Model probability: 62.7%

Class probabilities:

- Away Win: 36.7%
- Draw: 0.6%
- Home Win: 62.7%

This is a model prediction, not a certainty.

Annotation

The router identified the query as a prediction request and routed it to the prediction node. The system resolved the aliases “Pies” and “Cats” to the canonical dataset team names and called the match-winner prediction tool. The tool returned the predicted winner and probability. Validation confirmed that the required prediction fields were available, so the graph continued to the response formatter. The formatter presented the prediction with its probability and uncertainty disclaimer.

Trace 2: Retrieval

1. Query

What is Geelong's record against Essendon?

2. Router Decision

- **Intent:** retrieval
- **Route:** retrieval

3. Tool Called: `team_record_tool`

- **Input:**
 - `team_a` = Geelong Cats
 - `team_b` = Essendon Bombers
- Matches: 63
- Wins: 39
- Losses: 23
- Draws: 1

4. Validation

- **Status:** valid
- **Reason:** Tool result passed validation.

5. Final Response

Geelong Cats' record against Essendon Bombers is 39 wins, 23 losses, and 1 draw across 63 matches.

Annotation

This is an exact statistical/data question, so the router selected the retrieval path. The tool retrieved the numbers directly from the structured AFL dataset rather than asking the LLM to invent them. The retrieval result contained usable data, so the graph proceeded to the response formatter. The formatter converted the structured dictionary into a concise natural-language answer.

Trace 3: Ambiguous Prediction

1. Query

Who will win Tigers vs Unknown FC?

2. Router Decision

- **Intent:** prediction
- **Route:** prediction

3. Tool Called: `match_winner_prediction_tool`

- **Input:**
 - `team_a` = Richmond Tigers
 - `team_b` = Unknown FC

- Tigers → Richmond Tigers
- Unknown FC → unresolved
- *Error: “Could not resolve two AFL teams from the prediction request.”*

4. Validation

- **Status:** clarification
- **Reason:** Could not resolve two AFL teams from the prediction request.
- **Needs clarification:** True

5. Final Response

I couldn't resolve all the information needed for this request. Please provide a specific AFL team, player, or match date.

Annotation

The router identified the query as a prediction request and routed it to the prediction node. The system successfully resolved “Tigers” to “Richmond Tigers,” but “Unknown FC” could not be matched to an AFL team in the available dataset. Instead of guessing the missing team, the prediction tool returned an error. The validation node detected the unresolved information and routed the request to the clarification node, which asked the user to provide specific AFL information.

Routing Accuracy Evaluation

The router was evaluated using 15 test queries covering factual AFL questions, structured retrieval requests, prediction requests, and off-topic questions.

#	Query	Expected Intent	Predicted Intent	Correct
1	What is a behind in AFL?	factual	factual	Yes
2	What are the basic rules of AFL?	factual	factual	Yes
3	What is a mark in AFL?	factual	factual	Yes
4	What is a handball in AFL?	factual	factual	Yes
5	What is Geelong's record against Essendon?	retrieval	retrieval	Yes
6	How many wins does Geelong have against Essendon?	retrieval	retrieval	Yes
7	What are Geelong's player stats?	retrieval	retrieval	Yes
8	How many goals did a player score?	retrieval	retrieval	Yes
9	Who will win Pies vs Cats?	prediction	prediction	Yes
10	Who is likely to win Richmond vs Carlton?	prediction	prediction	Yes
11	Who will be the top player for Geelong?	prediction	prediction	Yes
12	Who will top-score in the match?	prediction	prediction	Yes
13	What is the capital of France?	off-topic	off-topic	Yes

#	Query	Expected Intent	Predicted Intent	Correct
14	What is today's weather?	off-topic	off-topic	Yes
15	Tell me a Python joke.	off-topic	off-topic	Yes

Routing Accuracy

- **Total test cases:** 15
- **Correctly routed:** 15
- **Incorrectly routed:** 0
- **Routing Accuracy:** 100%

Formula

Routing Accuracy = (Correctly Routed Queries ÷ Total Queries) × 100

Routing Accuracy = (15 ÷ 15) × 100 = 100%

Evaluation Summary

The router correctly distinguished between factual AFL questions, structured retrieval requests, prediction requests, and off-topic questions across the tested cases.